
LECTURE NOTES

Discrete Mathematics notes

Sequences

DM549 - Discrete Mathematics

Simon Holm
Fall 2024



DEPARTMENT OF MATHEMATICS
AND COMPUTER SCIENCE

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1 Sequences

Definition 2.4.1

A sequence (følge) is a function of $\mathbb{N}$ to some set.

Remarks:

- The domain of the function may be finite or infinite. Usually $\{0, 1, 2, \dots\}$ or $\{1, 2, 3, \dots\}$.
- Denote by a_n the number that the function maps to, also called a **term** of the sequence. (Recall our notation for the Fibonacci sequence!)
- The sequence is then denoted by $\{a_n\}$. Do not confuse this with set notation!
- We can also list the terms in order, e.g., for the domain being $\mathbb{N}$:

$$\{a_n\} = a_0, a_1, a_2, a_3, \dots$$

- You can also think of a sequence with domain D as a $|D|$ -tuple.

1.1 Examples of sequences of numbers

- Fibonacci sequence: $f_n = f_{n-1} + f_{n-2} = \{1, 1, 2, 3, 5, 8, \dots\}$
- All numbers for $\frac{1}{n}$ where n is greater than 1: $\{\frac{1}{n}\}_{n \geq 1} = \{0.5, 0.33, 0.25, \dots\}$

1.2 Defining geometric and arithmetic sequences

Definition

An infinite **geometric sequence** (geometrisk følge) is a sequence of the form

$$a_n = c \cdot r^n, \quad n \in \mathbb{N}$$

where $c \in \mathbb{R}$ is the **initial term** (begyndelsesled) and $r \in \mathbb{R}$ is the **common ratio** (fælles faktor). We obtain finite geometric sequences by stopping at some point.

Definition

An infinite **arithmetic sequence** (aritmetisk følge) is a sequence of the form

$$a_n = b + n \cdot d, \quad n \in \mathbb{N}$$

where $b \in \mathbb{R}$ is the **initial term** (begyndelsesled) and $d \in \mathbb{R}$ is the **common difference** (fælles differens). We obtain finite arithmetic sequences by stopping at some point.

1.3 Series (rækker, not “serier”)

Series

Let

$$a_m, a_{m+1}, \dots, a_n$$

be a sequence. Then there is an associated **series** (række), the sum of all terms in the sequence. It is denoted by

$$\sum_{i=m}^n a_i \quad \text{or} \quad \sum_{m \leq i \leq n} a_i$$

Remarks:

- If the domain of the sequence is D , we also write $\sum_{i \in D} a_i$.
- One could also talk about series that are the sum of infinitely many terms of an infinite sequence.
- Here, we focus on finite sequences.
- Otherwise, to be completely formal, we would need to talk about a concept from calculus called convergence.

1.3.1 Geometric series

Theorem 2.4.1

For a finite **geometric series** (with $c = 1$), the series corresponding to a finite geometric sequence, it holds that

$$\sum_{m \leq i \leq n} a_i = \begin{cases} \frac{r^{n+1}-1}{r-1} & \text{if } r \in \mathbb{R} \setminus \{1\} \\ n+1 & \text{if } r = 1. \end{cases}$$

Note: if $|r| < 1$ and we consider the **infinite** geometric series, the term r^{n+1} vanishes as n grows towards ∞ , so

$$\sum_{i=0}^{\infty} r^i = \frac{1}{1-r}$$

1.3.2 Arithmetic series

Theorem 2.4.1

For a finite **arithmetic series**, the series corresponding to a finite arithmetic sequence, it holds that

$$\sum_{i=0}^n (b + i \cdot d) = b \cdot (n + 1) + d \cdot \frac{n \cdot (n + 1)}{2}$$

Some useful sequences:

<i>n</i> th term	First 10 terms
n^2	{1, 4, 9, 16, 25, 36, 49, 64, 81, 100, ...}
n^3	{1, 8, 27, 64, 125, 216, 343, 512, 729, 1000, ...}
n^4	{1, 16, 81, 256, 625, 1296, 2401, 4096, 6561, 10000, ...}
2^n	{2, 4, 8, 16, 32, 64, 128, 256, 512, 1024, ...}
3^n	{3, 9, 27, 81, 243, 729, 2187, 6561, 19683, 59049, ...}
$n!$	{1, 2, 6, 24, 120, 720, 5040, 40320, 362880, 3628800, ...}
F_n (fibonacci)	{1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, ...}

1.4 Proofs

Example of a geometric sequence:

$$1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \dots$$

and the corresponding geometric series would be:

$$1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots$$

Recall:

$$\sum_{i=0}^n \left(\frac{1}{2}\right)^i = 2 - \left(\frac{1}{2}\right)^n$$

and we have that:

$$\sum_{i=0}^{\infty} \left(\frac{1}{2}\right)^i = \frac{1}{1 - \frac{1}{2}} = 2$$

□

1.4.1 Proof of the finite geometric series formula

Let (geometric series definition):

$$S = \sum_{i=0}^n r^i$$

And observe that:

$$r \cdot S = \sum_{i=0}^{n+1} r^i = \sum_{i=0}^n r^i + r^{n+1} - 1$$

When solving for S , we get

$$S = \frac{r^{n+1} - 1}{r - 1} \quad \text{when } r \neq 1$$

For $r = 1$, it is clear that

$$S = \sum_{i=0}^n r^i = n + 1$$

□

1.4.2 Proof of the finite arithmetic series formula

$$\begin{aligned} \sum_{i=0}^n (b + i \cdot d) &= \sum_{i=0}^n b + \sum_{i=0}^n i \cdot d \\ &= b \cdot (n + 1) + d \cdot \sum_{i=0}^n i \\ &= b \cdot (n + 1) + d \cdot \frac{n \cdot (n + 1)}{2}. \end{aligned}$$

□